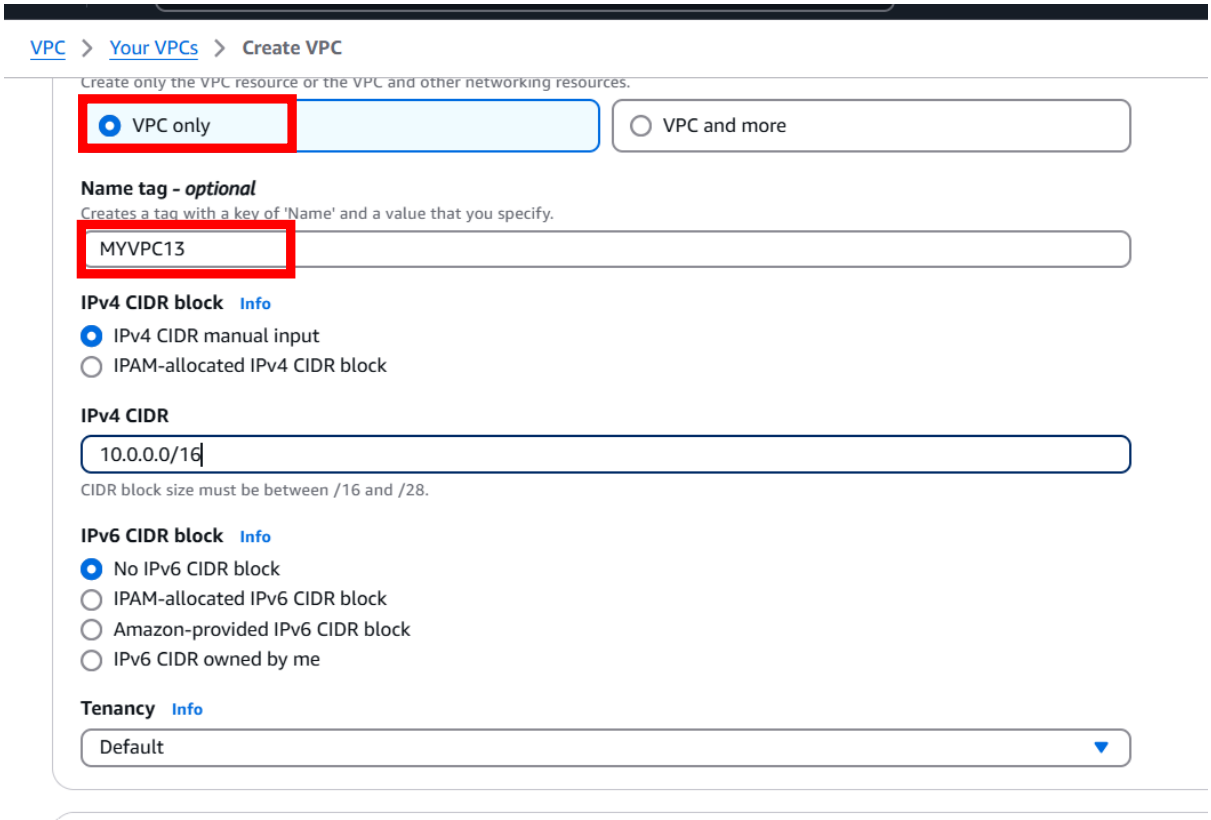
CREATION OF VPC-LINUX

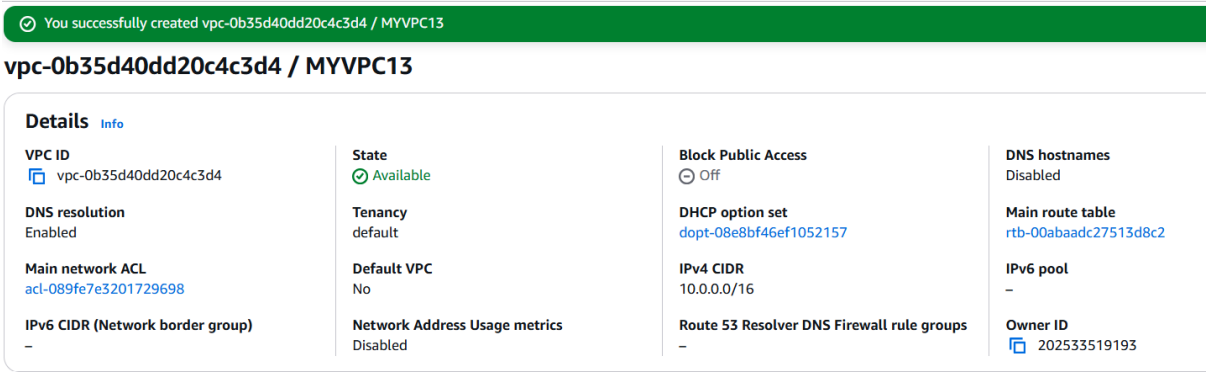
Step 1: Creation of VPC

Find vpc in AWS console -> Select VPC ->Create VPC

1) Select VPC only
2) Name tag as MYVPC13
3) Selected IPV4 CIDR manual input
4) Given CIDR as 10.0.0.0/16
5) Select No IPV6
6) Click on create VPC



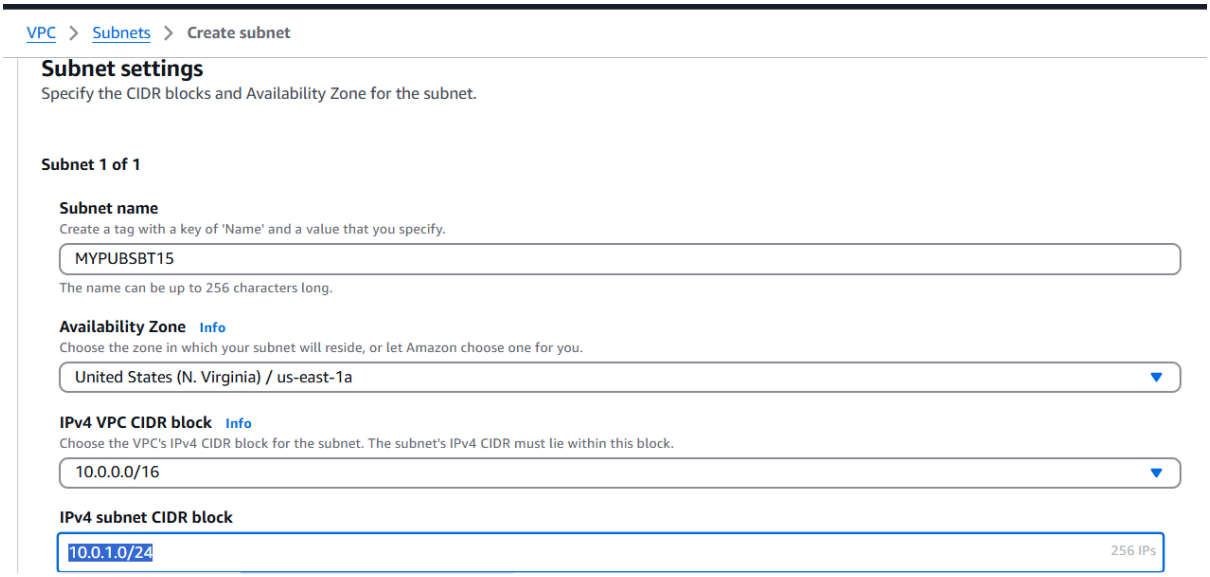
1. VPC is successfully created



Step 2: Creation of subnets for public and private with different available zones by assigning different CIDR blocks

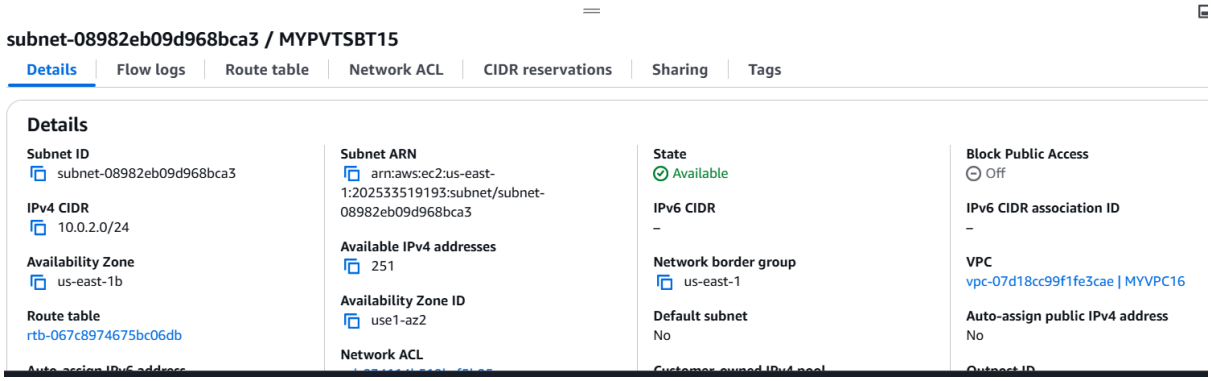
Creation of PUBLIC subnet

1. **Selected the VPC ID -which is the VPC created in the previous step**
2. **Edited the subnet settings**
3. **Chosen availability zone**
4. **Typed IPv4 subnet CIDR block as 10.0.1.0/24**
5. **Select ->create a subnet**

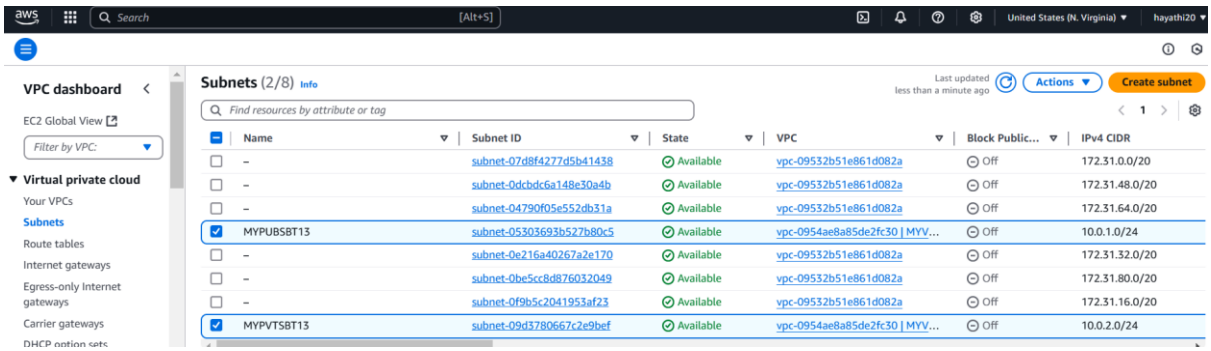


Creation of PRIVATE subnet

1. **Selected the VPC ID -which is the VPC created in the previous step**
2. **Edited the subnet settings**
3. **Chosen the different availability zone**
4. **Typed IPv4 subnet CIDR block as 10.0.2.0/24**
5. **Select ->create a subnet**



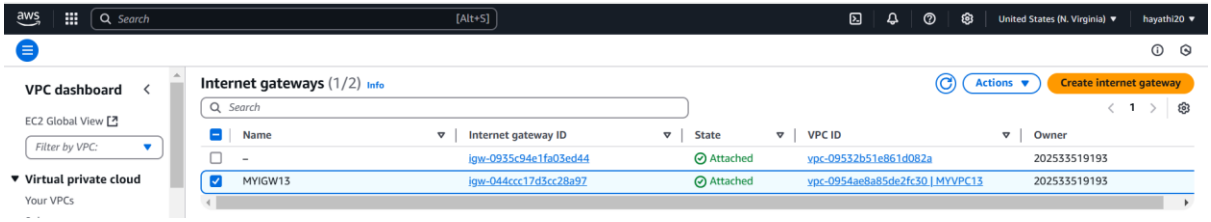
Below, two subnets have been created with two different zones.



Step 3: Create Internet gateway and attaching with VPC.

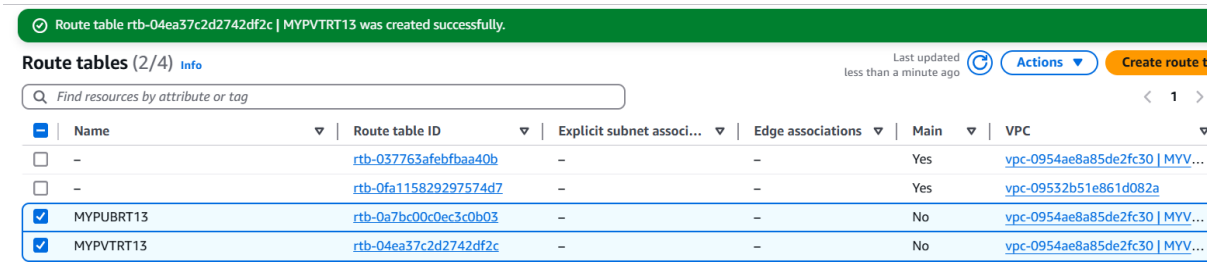
Click Internet gateway-> Create Internet gateway

Click on attach to VPC-> select attach

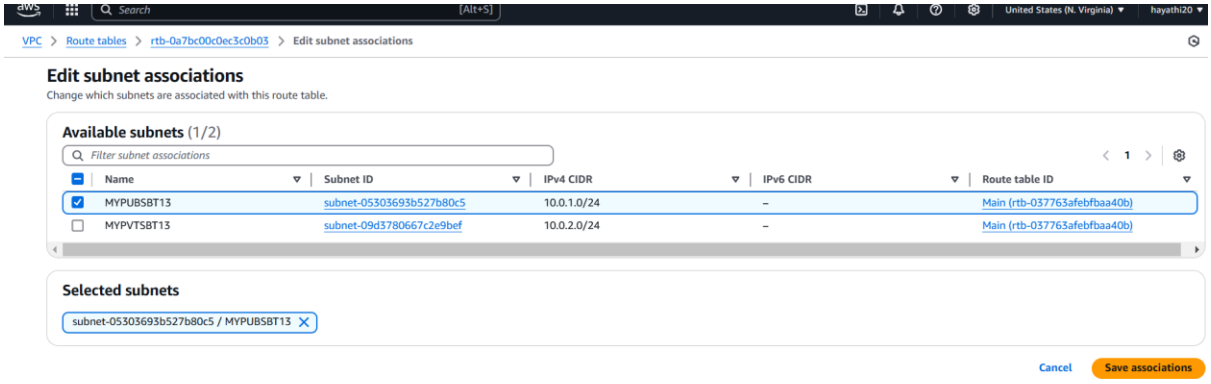


Step 4: Creation of Route table one as public and one as private, associate with subnets

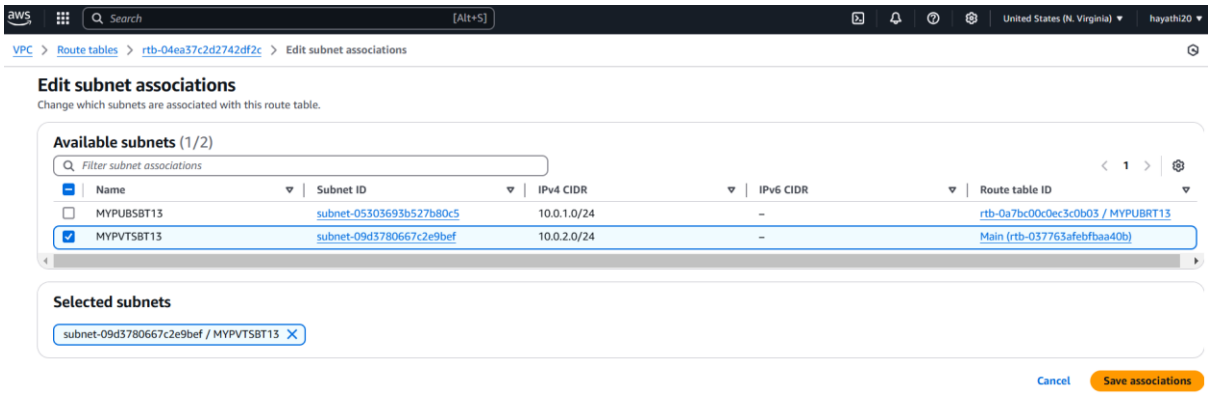
A private and public routing table is created and shown below.



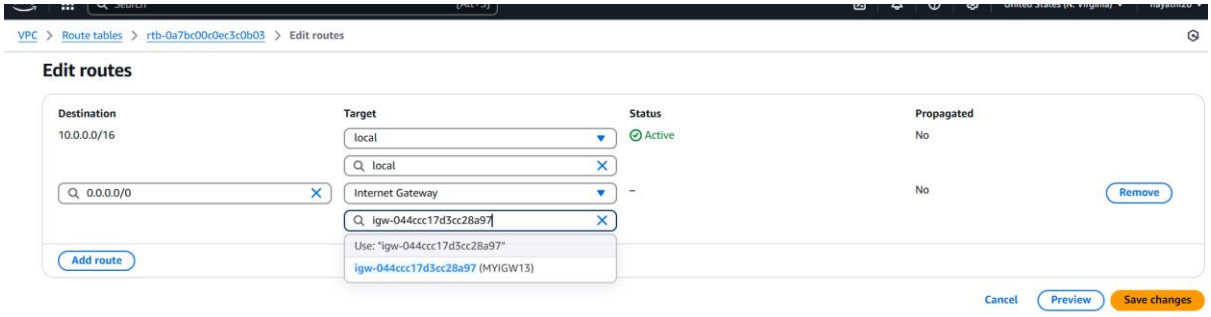
Associating public routing table with public subnet



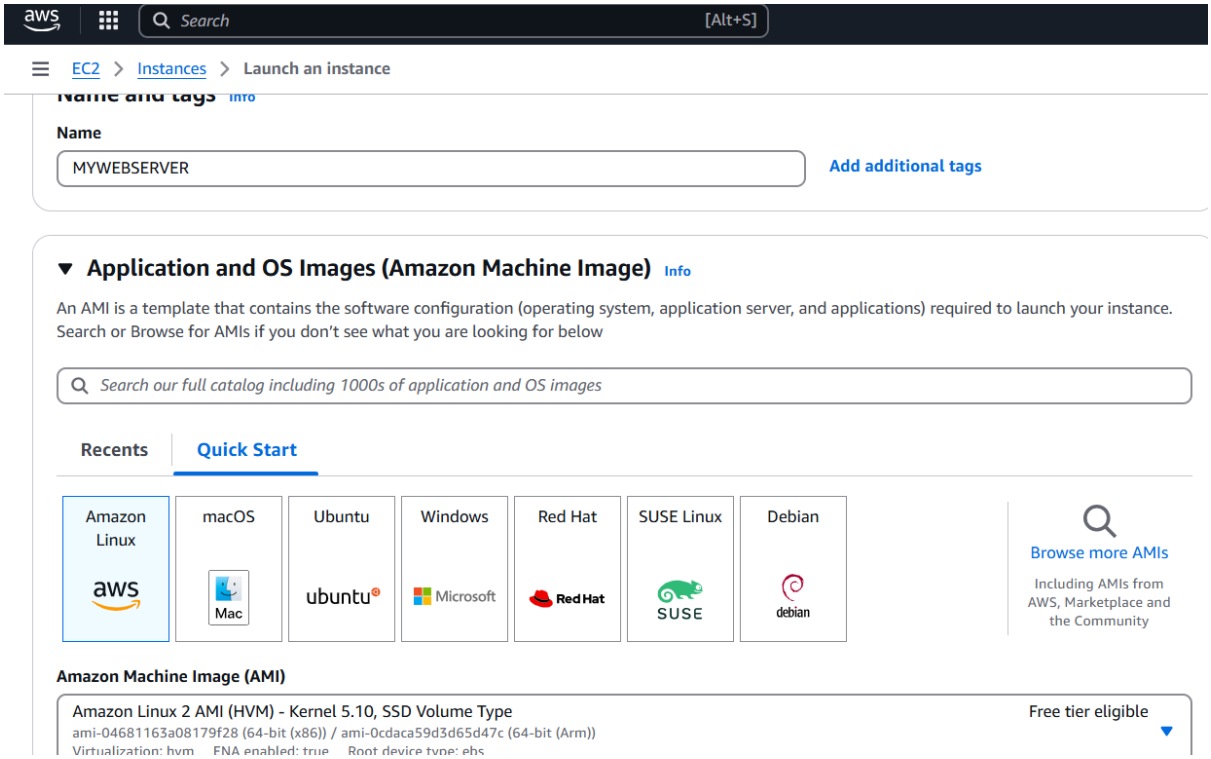
Associating private routing table with private subnet

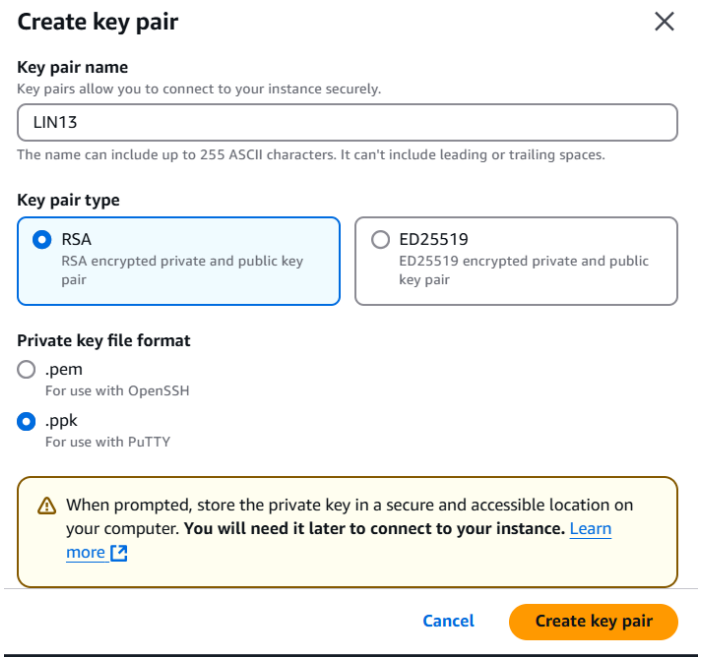


Step 5: Edit the public route table Route alone and map the IGW, not the private



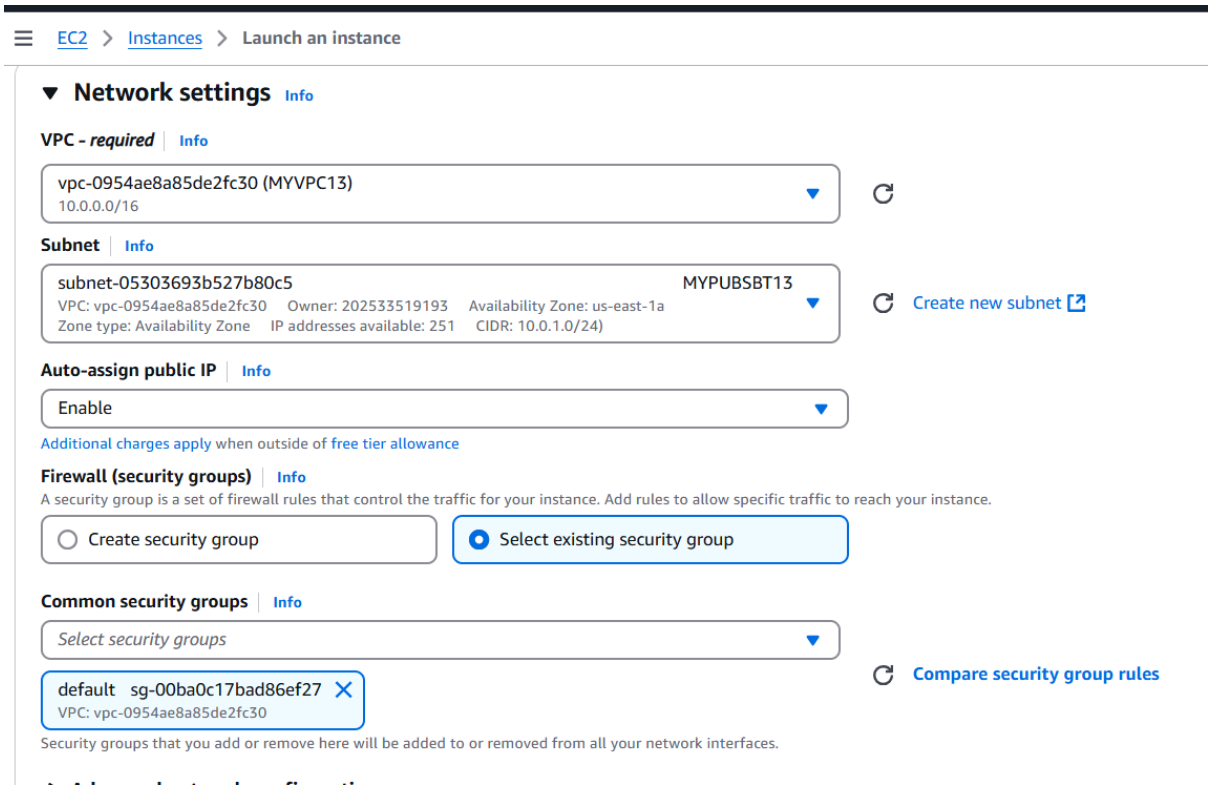
Step 6: Creation of two EC2 instances-Linux



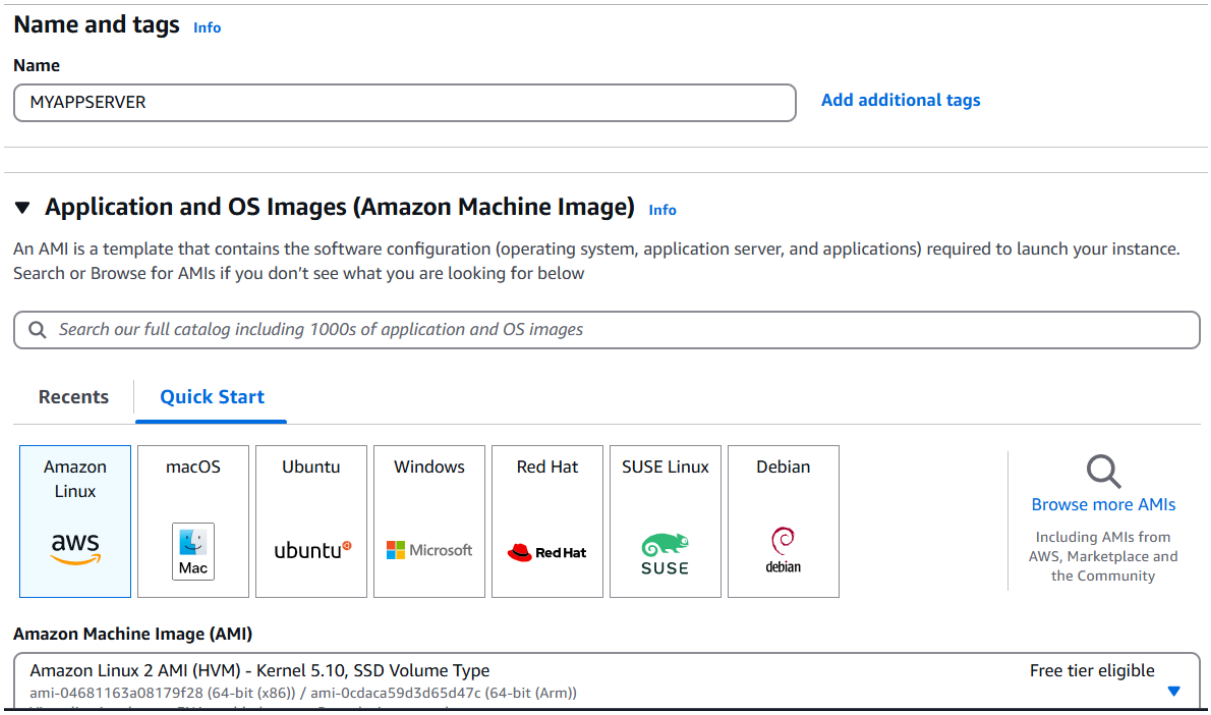


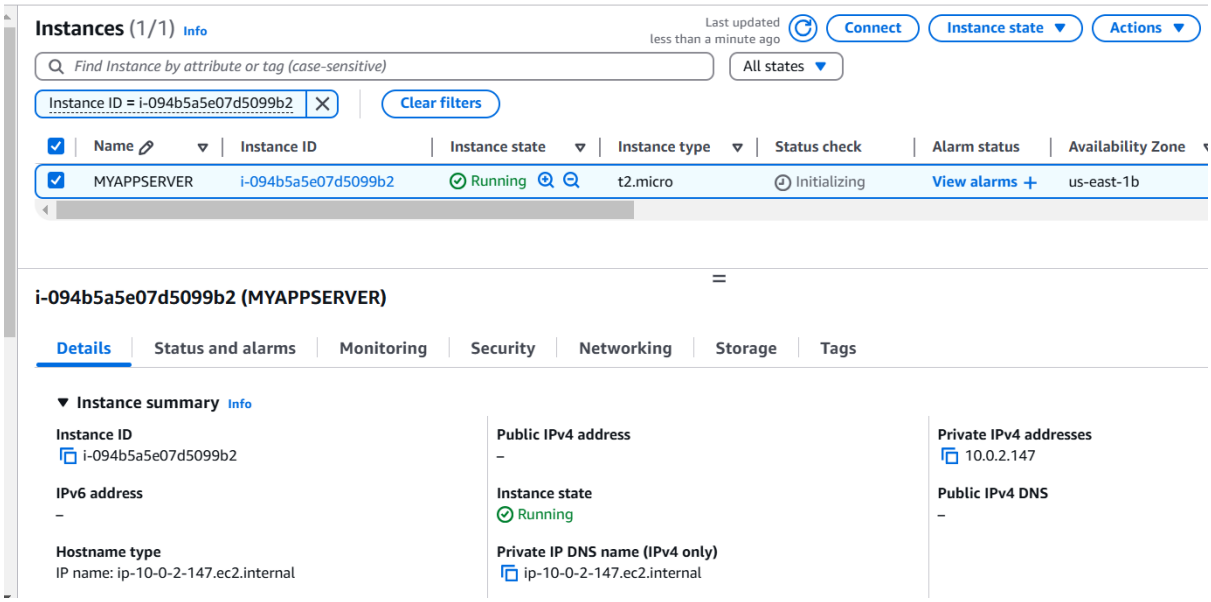
Step 6 : Create two security groups

Edit in Network settings and setup is done as shown below, it is for public



Step 7 : Launching EC2 instance as MYAPPSERVER -PRIVATE



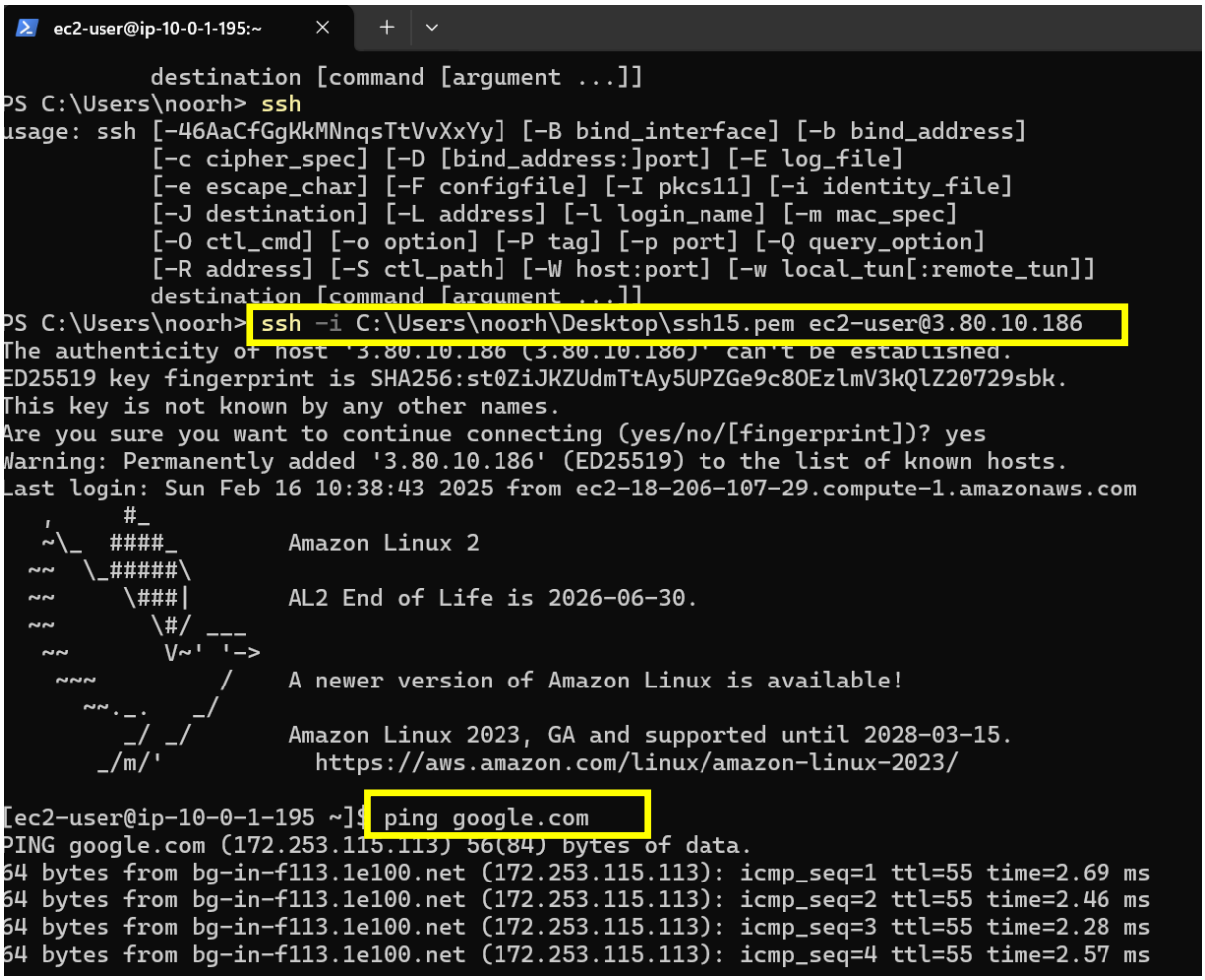


Step 8: Login to public and check for internet connection

Click on Windows-> powershell

Typed the command as `ssh -i C:\Users\noorh\Desktop\ssh15.pem ec2-user@3.80.10.186`

Now logged into amazon Linux server

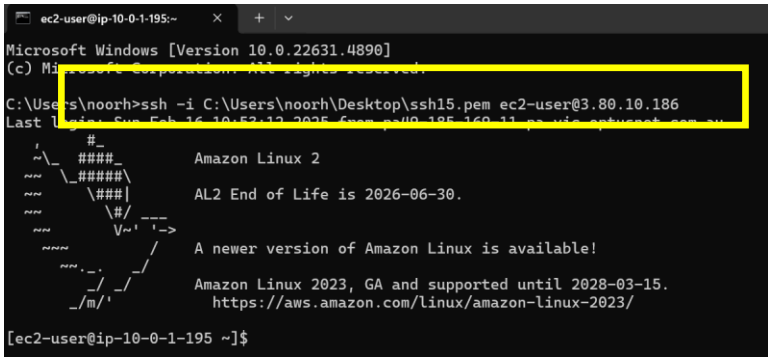


Checked the connection by giving the command as `ping google.com`

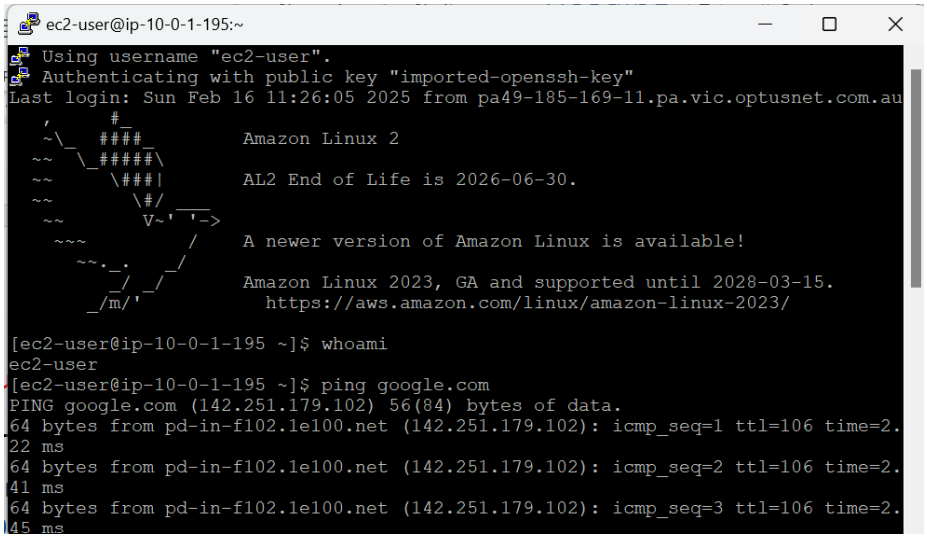
and got the below output.

```
ec2-user@ip-10-0-1-195 ~]$ ping google.com
ING google.com (172.253.115.113) 56(84) bytes of data.
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=1 ttl=55 time=2.69 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=2 ttl=55 time=2.46 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=3 ttl=55 time=2.28 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=4 ttl=55 time=2.57 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=5 ttl=55 time=2.54 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=6 ttl=55 time=2.28 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=7 ttl=55 time=2.57 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=8 ttl=55 time=2.84 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=9 ttl=55 time=2.69 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=10 ttl=55 time=2.28 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=11 ttl=55 time=2.45 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=12 ttl=55 time=2.28 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=13 ttl=55 time=2.45 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=14 ttl=55 time=2.41 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=15 ttl=55 time=2.46 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=16 ttl=55 time=2.59 ms
4 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=17 ttl=55 time=2.66 ms
C
-- google.com ping statistics ---
7 packets transmitted, 17 received, 0% packet loss, time 16026ms
tt min/avg/max/mdev = 2.282/2.504/2.847/0.175 ms
```

The same thing is cross-verified in the local system command prompt and successfully got the server running.



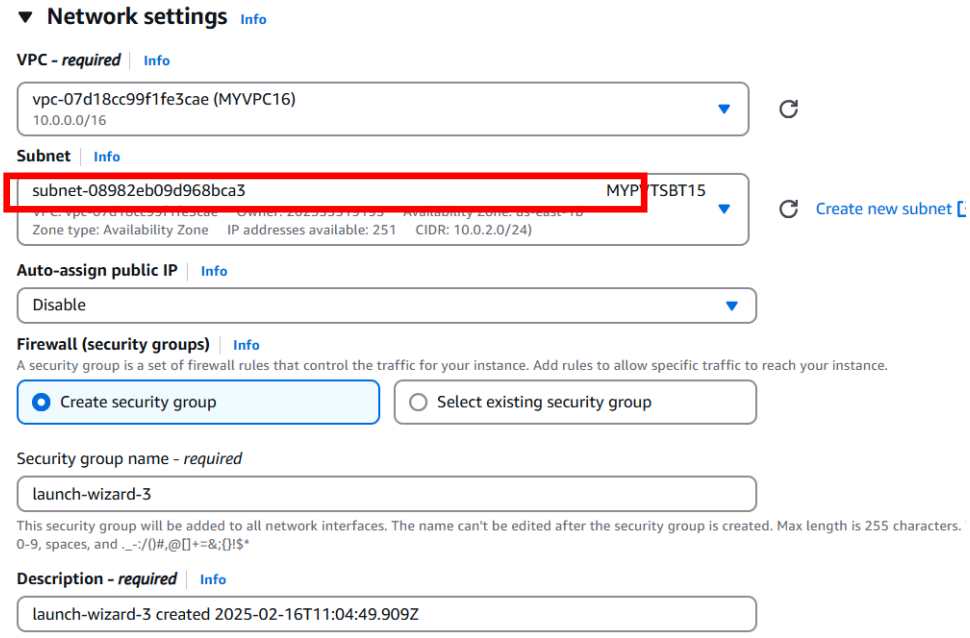
Putty output:



Creation of Private instances:

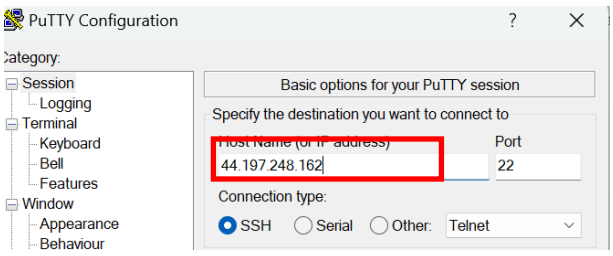
EC2->Launch Instances

1) Named as appserver
2) Selected AMI as Amazon linux
3) Instance type as t2.micro
4) Key pair is selected from the existing keypair
5) Edit the network settings
6) Disable the Auto-assign public IP

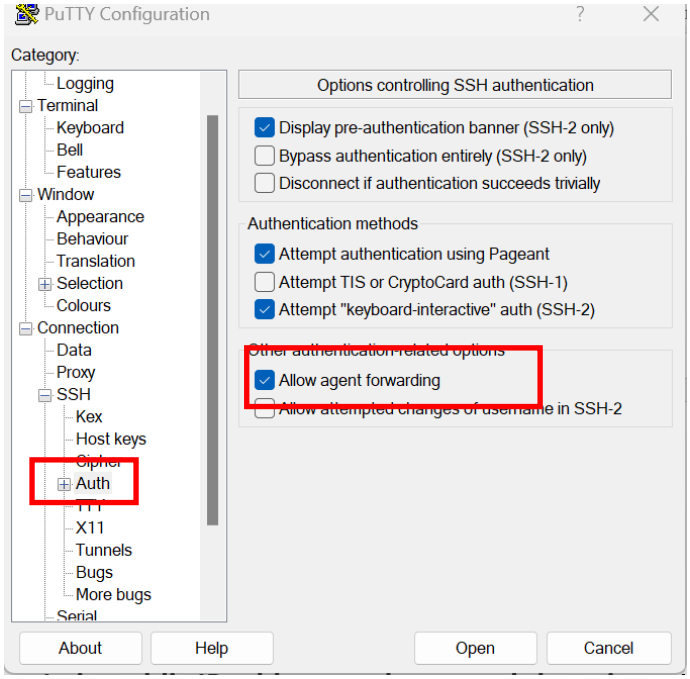


Launched the putty - Logged into putty in public IP address and accessed the private IP

Host name : Public IP



In SSH-> Auth -> Enable agent forwarding



After connecting to public IP it has been connected to private IP .

Given command: ssh ec2-user@Private IP address

